Nashville Housing – Data Cleaning with SQL Project

Workbook & Thought Process

Data File: Nashville-Housing-Data.xlsx

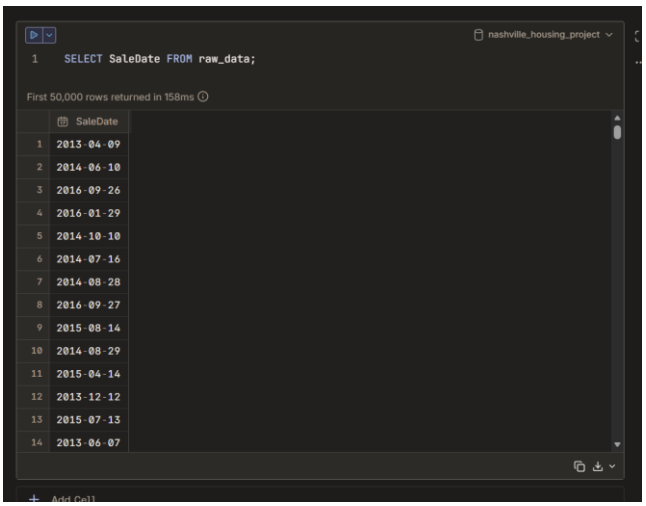
Data Ingestion:



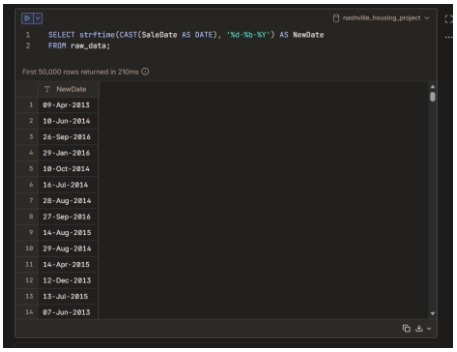
Data Cleaning:

-Standardize Date Format:

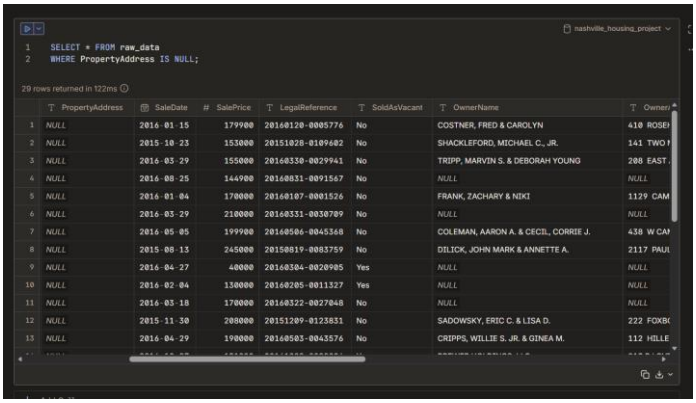
I am checking the date format to solve the possible confusions about the date format (such as American vs. European format confusion).



Although the dates are in the YYYY-MM-DD format, I will convert them to the DD-MMM-YYYY format (e.g., 09-Apr-2013) to improve readability.



-Populate the Property Address:

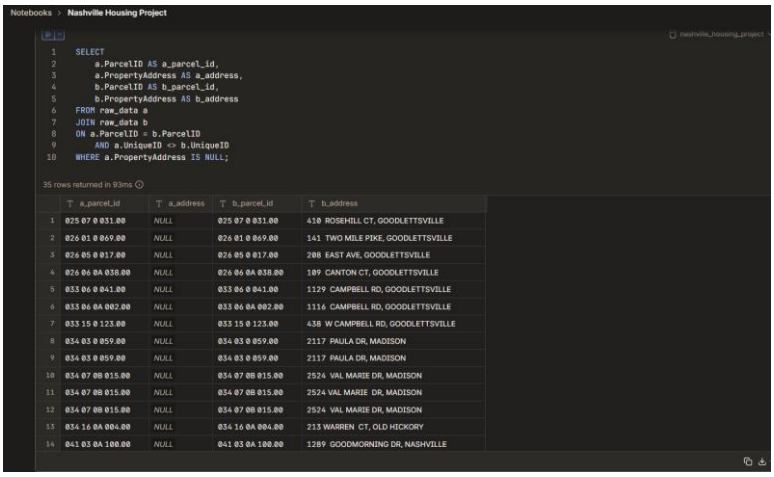


```
SELECT * FROM raw_data
WHERE PropertyAddress IS NULL;
```

29 rows returned in 122ms

	PropertyAddress	SaleDate	SalePrice	LegalReference	SoldAsVacant	OwnerName	Owner
1	NULL	2016-01-15	179900	20160120-0005776	No	COSTNER, FRED & CAROLYN	410 ROSE
2	NULL	2015-10-23	153000	20151028-0109602	No	SHACKLEFORD, MICHAEL C., JR.	141 TWO
3	NULL	2016-03-29	155000	20160330-0029941	No	TRIPP, MARVIN S. & DEBORAH YOUNG	208 EAST
4	NULL	2016-08-25	144900	20160831-0091567	No	NULL	NULL
5	NULL	2016-01-04	170000	20160107-0001526	No	FRANK, ZACHARY & NIKI	1129 CAM
6	NULL	2016-03-29	210000	20160331-0030709	No	NULL	NULL
7	NULL	2016-05-05	199900	20160506-0045368	No	COLEMAN, AARON A. & CECIL, CORRIE J.	438 W CA
8	NULL	2015-08-13	245000	20150819-0083759	No	DELICK, JOHN MARK & ANNETTE A.	2117 PAUL
9	NULL	2016-04-27	40000	20160304-0020905	Yes	NULL	NULL
10	NULL	2016-02-04	130000	20160205-0011327	Yes	NULL	NULL
11	NULL	2016-03-18	170000	20160322-0027048	No	NULL	NULL
12	NULL	2015-11-30	208000	20151209-0123831	No	SADOWSKY, ERIC C. & LISA D.	222 FOXB
13	NULL	2016-04-29	190000	20160503-0043576	No	CRIPPS, WILLIE S. JR. & GINEA M.	112 HILLE

By performing a self-join operation on the dataset based on matching parcel IDs, I will find and populate the missing property addresses.



Notebooks › Nashville Housing Project

```
SELECT
    a.ParcelID AS a_parcel_id,
    a.PropertyAddress AS a_address,
    b.ParcelID AS b_parcel_id,
    b.PropertyAddress AS b_address
FROM raw_data a
JOIN raw_data b
ON a.ParcelID = b.ParcelID
    AND a.UniqueID <> b.UniqueID
    WHERE a.PropertyAddress IS NULL;
```

35 rows returned in 93ms

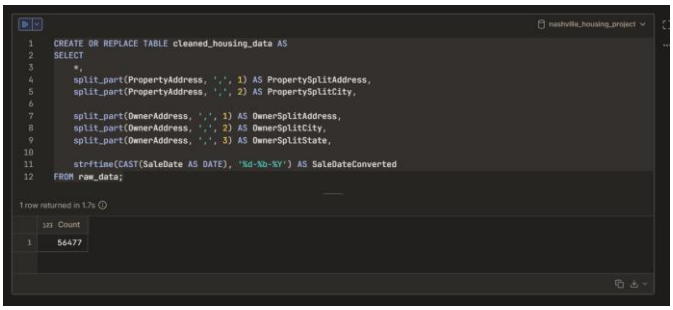
	a_parcel_id	a_address	b_parcel_id	b_address
1	025 07 0 031.00	NULL	025 07 0 031.00	410 ROSEHILL CT, GOODLETTSVILLE
2	026 01 0 069.00	NULL	026 01 0 069.00	141 TWO MILE PIKE, GOODLETTSVILLE
3	026 05 0 017.00	NULL	026 05 0 017.00	208 EAST AVE, GOODLETTSVILLE
4	026 06 0A 038.00	NULL	026 06 0A 038.00	109 CANTON CT, GOODLETTSVILLE
5	033 06 0 041.00	NULL	033 06 0 041.00	1129 CAMPBELL RD, GOODLETTSVILLE
6	033 06 0A 002.00	NULL	033 06 0A 002.00	1116 CAMPBELL RD, GOODLETTSVILLE
7	033 15 0 123.00	NULL	033 15 0 123.00	438 W CAMPBELL RD, GOODLETTSVILLE
8	034 03 0 059.00	NULL	034 03 0 059.00	2117 PAULA DR, MADISON
9	034 03 0 059.00	NULL	034 03 0 059.00	2117 PAULA DR, MADISON
10	034 07 0B 015.00	NULL	034 07 0B 015.00	2524 VAL MARIE DR, MADISON
11	034 07 0B 015.00	NULL	034 07 0B 015.00	2524 VAL MARIE DR, MADISON
12	034 07 0B 015.00	NULL	034 07 0B 015.00	2524 VAL MARIE DR, MADISON
13	034 16 0A 004.00	NULL	034 16 0A 004.00	213 WARREN CT, OLD HICKORY
14	041 03 0A 100.00	NULL	041 03 0A 100.00	1289 GOODMORNING DR, NASHVILLE



```
UPDATE raw_data AS a
SET PropertyAddress=b.PropertyAddress
FROM raw_data AS b
WHERE a.ParcelID = b.ParcelID
    AND a.UniqueID <> b.UniqueID
    AND a.PropertyAddress IS NULL;
SELECT PropertyAddress FROM raw_data;
```

First 50,000 rows returned, 2 statements run in 427ms

-Separating address into individual columns (Address, City, State) and creating a new table with date format and split addresses:



```
CREATE OR REPLACE TABLE cleaned_housing_data AS
SELECT
    *,
    split_part(PropertyAddress, ',', 1) AS PropertySplitAddress,
    split_part(PropertyAddress, ',', 2) AS PropertySplitCity,

    split_part(OwnerAddress, ',', 1) AS OwnerSplitAddress,
    split_part(OwnerAddress, ',', 2) AS OwnerSplitCity,
    split_part(OwnerAddress, ',', 3) AS OwnerSplitState,

    strftime(CAST(SaleDate AS DATE), '%d-%b-%Y') AS SaleDateConverted
FROM raw_data;
```

1 row returned in 1.7s

	Count
1	56477

To improve data readability and consistency for future analysis, I separated the combined ‘PropertyAddress’ and ‘OwnerAddress’ columns into individual components (Address, City, State).

-Fixing Inconsistencies:

- Yes, No, or Y in the ‘SoldAsVacant’ column :



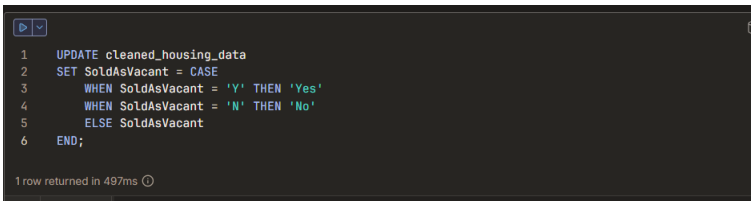
```
SELECT DISTINCT(SoldAsVacant) FROM cleaned_housing_data;
```

4 rows returned in 180ms

	SoldAsVacant
1	Yes
2	No
3	Y
4	N

While exploring the dataset, I noticed inconsistencies in the ‘SoldAsVacant’ column. The column contained a mix of 'Yes/No' and 'Y/N' values.

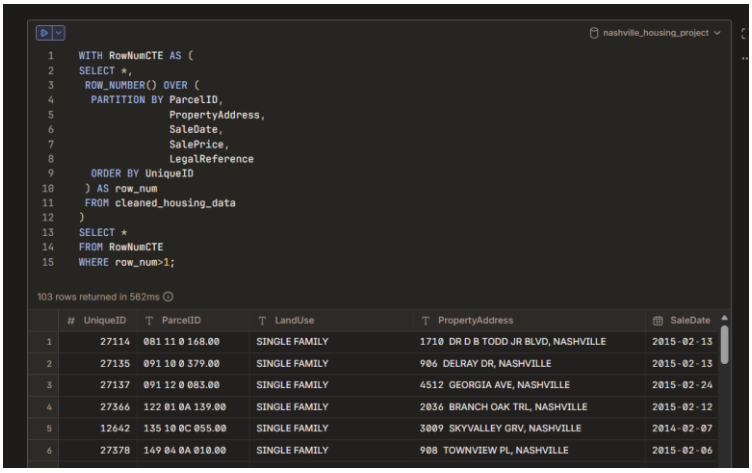
To ensure data consistency, I wrote a CASE statement to standardize all 'Y' values to 'Yes' and 'N' values to 'No'. This step is crucial for accurate grouping and filtering during the Exploratory Data Analysis (EDA) phase.



```
UPDATE cleaned_housing_data
SET SoldAsVacant = CASE
    WHEN SoldAsVacant = 'Y' THEN 'Yes'
    WHEN SoldAsVacant = 'N' THEN 'No'
    ELSE SoldAsVacant
END;
```

1 row returned in 497ms

-Removing Duplicates:



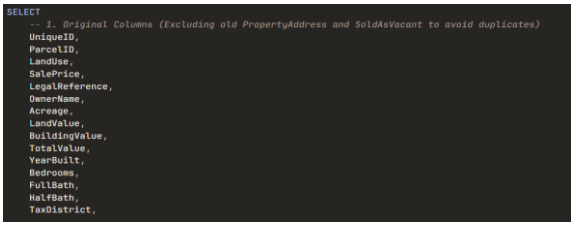
```
WITH RowNumCTE AS (
SELECT *,
 ROW_NUMBER() OVER (
  PARTITION BY ParcelID,
               PropertyAddress,
               SaleDate,
               SalePrice,
               LegalReference
  ORDER BY UniqueID
  ) AS row_num
 FROM cleaned_housing_data
)
SELECT *
FROM RowNumCTE
WHERE row_num>1;
```

103 rows returned in 562ms

	UniqueID	ParcelID	LandUse	PropertyAddress	SaleDate
1	27114	081 11 0 168.00	SINGLE FAMILY	1710 DR D B TODD JR BLVD, NASHVILLE	2015-02-13
2	27135	091 10 0 379.00	SINGLE FAMILY	906 DELRAY DR, NASHVILLE	2015-02-13
3	27137	091 12 0 083.00	SINGLE FAMILY	4512 GEORGIA AVE, NASHVILLE	2015-02-24
4	27366	122 01 0A 139.00	SINGLE FAMILY	2036 BRANCH OAK TRL, NASHVILLE	2015-02-12
5	12642	135 10 0C 055.00	SINGLE FAMILY	3009 SKYVALLEY GRV, NASHVILLE	2014-02-07
6	27378	149 04 0A 010.00	SINGLE FAMILY	908 TOWNVIEW PL, NASHVILLE	2015-02-06

I used a CTE (Common Table Expression) with the ROW_NUMBER() window function to identify duplicate records. By partitioning the data by key identifiers (ParcelID, Address, SaleDate, etc.) and ordering by UniqueID, I discovered 103 duplicate rows in the dataset that needed to be addressed.

-Choosing Only Necessary Columns:



```
SELECT
    -- 1. Original Columns (Excluding old PropertyAddress and SoldAsVacant to avoid duplicates)
    UniqueID,
    ParcelID,
    LandUse,
    SalePrice,
    LegalReference,
    OwnerName,
    Acreage,
    LandValue,
    BuildingValue,
    TotalValue,
    YearBuilt,
    Bedrooms,
    FullBath,
    HalfBath,
    TaxDistrict,
```

Final Cleaned Table: Summary & Methodology

Overview:
After performing exploratory data analysis (EDA) and testing various transformations on the raw data, I successfully generated the final cleaned_housing_data table. As seen in the screenshots, the dataset is now fully standardized, free of duplicate records, and optimized for analysis.

Summary of Actions Taken:

1. **Idempotent Pipeline (Repeatability):** Instead of modifying the raw table using destructive UPDATE or DELETE statements, I put all transformations into a single, repeatable CREATE OR REPLACE TABLE query. (In data_cleaning_pipeline.sql file). This ensures the pipeline can be run multiple times without errors and keeps the raw data immutable.
2. **Address Standardization:** Split the combined PropertyAddress and OwnerAddress into individual, readable columns (Address, City, State). I also used window functions to populate missing property addresses before splitting them.
3. **Date Formatting:** Converted the raw date into a standard DD-MMM-YYYY format (e.g., 18-Jul-2014) to improve readability and prevent international date format confusion.
4. **Data Consistency:** Standardized the SoldAsVacant column, converting all 'Y' and 'N' values to a consistent 'Yes' and 'No' format.
5. **Duplicate Removal:** Utilized QUALIFY ROW_NUMBER() partitioned by key identifiers (ParcelID, Address, SaleDate, SalePrice, LegalReference) to ensure that only unique property records are ingested into the final table.
6. **Column Optimization:** Removed unused or redundant columns (like the original combined addresses and raw dates) to keep the final table clean, performant, and easy to query.

Conclusion:
The data is now highly reliable, structured, and ready for Exploratory Data Analysis (EDA) and visualization!